

CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY



EXPERIMENT 10

**Subject: Advanced Database Management
System**

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Question 1:

-----Hard Level Problem-----

/*

Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction.

If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints.

The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

HINT: YOU HAVE TO USE SAVEPOINTS

*/

Create a procedure:

```
395
396 DO $$
397 BEGIN
398
399 BEGIN
400     -- Insert multiple students
401     INSERT INTO students(name, age, class) VALUES ('Anisha',16,8);
402     INSERT INTO students(name, age, class) VALUES ('Neha',17,8);
403     INSERT INTO students(name, age, class) VALUES ('Mayank',19,9);
404
405     -- If all succeed
406     RAISE NOTICE ' Transaction Successfully Done';
407
408 EXCEPTION WHEN OTHERS THEN
409     -- If any insert fails
410     RAISE NOTICE 'Transaction Failed..! Rolling back changes.';
411     RAISE; -- this will rollback the entire transaction
412 END;
413 END;
414 $$;
415 |
416
```

Data Output Messages Notifications

NOTICE: Transaction Successfully Done
DO

Query returned successfully in 57 msec.

```
415
416
417 SELECT * FROM students;
418
```

Data Output Messages Notifications

	id [PK] integer	name character varying (50)	age integer	class integer
1	1	Anisha	16	8
2	2	Neha	17	8
3	3	Mayank	19	9

Showing rollback in case of exception:

```
420
421 --VIOLATED SCENARIO
422 DO $$
423 BEGIN
424     -- Start a transaction
425     BEGIN
426         -- Insert multiple students
427         INSERT INTO students(name, age, class) VALUES ('Anisha',16,8);
428         INSERT INTO students(name, age, class) VALUES ('Mayank',19,9);
429         INSERT INTO students(name, age, class) VALUES ('Anisha',17,8); --ERROR
430         INSERT INTO students(name, age, class) VALUES ('Mayank',19,9);
431
432         -- If all succeed
433         RAISE NOTICE ' Transaction Successfully Done';
434
435     EXCEPTION WHEN OTHERS THEN
436         -- If any insert fails
437         RAISE NOTICE 'Transaction Failed..! Rolling back changes.';
438         RAISE; -- this will rollback the entire transaction
439     END;
440 END;
441 $$;
442
```

Data Output Messages Notifications

NOTICE: Transaction Failed..! Rolling back changes.

ERROR: duplicate key value violates unique constraint "students_name_key"
Key (name)=(Anisha) already exists.

SQL state: 23505
Detail: Key (name)=(Anisha) already exists.
Context: SQL statement "INSERT INTO students(name, age, class) VALUES ('Anisha',16,8)"
PL/pgSQL function inline_code_block line 6 at SQL statement